

Econometric game 2025

Anastasija Tetereva, Onno Kleen, Alla Petukhina

9-10 April 2025

1 Case A

Due to the increasing integration of photovoltaic (PV) and wind power into the German electricity grid, stable grid operation is currently maintained through congestion management or redispatch. The primary aim of this approach is to minimize the curtailment of renewable energy feed-in while utilizing flexibility within the electricity network. This process incurs significant costs, with national congestion management expenses now exceeding 1.4 billion euros, which are passed on to end consumers through grid service charges. The responsibility for grid stability lies with the transmission system operators (TSOs), who ensure the integration of distribution grids into the congestion management framework via a coordination cascade.

Analyzing congestion management patterns is essential to understanding its drivers and developing predictive models that can forecast potential bottlenecks, allowing for preemptive redispatch actions. Enhanced forecasting accuracy in congestion management can contribute to reducing associated costs by enabling targeted mitigation measures. In constructing a predictive model for such events, it is crucial to examine the relationship between congestion management and relevant exogenous variables, such as electricity demand (affected by holidays and workday hours), electricity supply (impacted by high or low wind and PV feed-in), and electricity prices. It should be noted, however, that intraday electricity prices may also be influenced by grid bottlenecks, meaning they are not entirely exogenous to the system.

You are provided with a prepared dataset containing redispatch measures along with potential forecasting model features. The data includes three main groups:

- Weather Data:
 - hourly indices for solar radiation,
 - air temperature,
 - and wind speed.
- Electricity Data:

- measures of electricity generation,
- consumption,
- and associated prices.
- Market Data:
 - prices of oil and gas,
 - carbon emissions,
 - indicators for working days and holidays.

These datasets will serve as inputs for building a forecasting model to predict redispatch events.

- Visualize the time series data that will be used for prediction, along with their relationships to key factors such as weather features, electricity price features, electricity generation and consumption forecasts, and natural fuel market features. Overlay or create side-by-side charts for the related features. Create scatter plots or correlation matrices to show the relationship between the predicted time series and the related features. Highlight any visible trends or correlations between the time series and related features. Offer insights into which factors seem to have the most significant impact on the target prediction.
- The primary objective is to develop a day-ahead predictive model to estimate the probability of congestion events within the German electricity network, in particular, for TenneT DE network. This model should operate on an hourly basis in Central European Time (CET) and provide the **likelihood of congestion (redispatch) occurrence of such events**. The prediction should be made for both directional flows together as well as separately for each hour of the next 24 hours.
- Given the unbalanced nature of redispatch data, what approaches would you take in model selection to address this imbalance?
- Visualize the historical positive and negative redispatch measures in relation to key influencing factors such as day-ahead and intraday prices, PV and wind power generation, and load curves. Use a geographical plot if it helps to effectively demonstrate these relationships.
- In your modeling approach, do you consider it more appropriate to model positive and negative congestion management separately? If so, what are your reasons for this decision?
- What factors influenced your choice of model? Considerations might include model suitability, training efficiency, and the avoidance of overly complex models with extended training times.

- What specific variables (features), assumptions, and simplifications did you incorporate in your model?
- In designing an adequate day-ahead forecast scenario, what training and forecasting routines did you adopt? For training purposes, consider using data from the period spanning January 1, 2020, to June 30, 2022. Use data from July 1, 2022, to December 31, 2022, to validate your model. Apply data from 2023 to evaluate the model's performance comprehensively.
- What specific steps would you take to ensure robust validation?
- What key metrics or indicators would you focus on in this evaluation?
- Describe your approach to backtesting. Would you implement an expanding or a rolling window technique, and what are the advantages and potential limitations of your chosen method?
- How would you assess the model's predictive accuracy? What insights could you provide based on this evaluation?
- Using visualizations, identify and discuss specific periods when your model performed well and when it struggled. What underlying factors might explain these variations in performance, and how could the model be further refined based on these observations?
- Provide policy recommendations based on your results.
- If there are several observations with the same timestamp, group them by hour.

2 Instructions for structuring the report

Your report should be clear, structured, and well-argued, demonstrating both methodological rigor and economic intuition. Below is the recommended structure:

1. Introduction
Briefly introduce the problem and its economic relevance. State your research question and objectives. Outline the structure of the report.
2. Data Description
Describe the dataset used, including sources, sample size, and time period. Explain key variables, transformations, and any data cleaning steps. Provide summary statistics and visualizations where relevant.
3. Methodology
Clearly explain the econometric model(s) applied. Justify the choice of estimation techniques. Discuss any assumptions and potential econometric issues.

4. Empirical Results
Present key estimation results in tables and figures. Interpret the coefficients and discuss statistical significance. Perform robustness checks and discuss potential limitations.
5. Discussion and Economic Interpretation
Relate your findings to economic theory and real-world implications. Compare results with existing literature. Discuss policy recommendations or business implications if relevant.
6. Conclusion
Summarize key findings concisely. Highlight limitations and potential extensions for future research.
7. References
List all academic papers, datasets, and other sources cited. Use a consistent citation format.
8. Appendix (if necessary)
Include additional tables, robustness tests, or technical derivations that support the main text.
9. The total volume of the document including Appendix should not exceed 20 pages. Be mindful of page formatting: use standard font size (e.g., 12 pt), appropriate margins (e.g., 1 inch), and spacing (e.g., 1.5 or 2x line spacing).

3 Instructions for submitting the report

- You can send any questions about the case or the competition to econometricgame@vsae.nl.
- You are not allowed to work on the report outside the competition hours. That is why we ask you to submit the (working) report on Wednesday before 18:00 by sending it to juryeg@vsae.nl. Title this email “First Day Case A - Team [Number]”. Use the number that was communicated to you in the email.
- The deadline for the final report submission is on Thursday at 17:00. You should do so by sending it to juryeg@vsae.nl. Please put your team number on the title page of your report. Do not include your names and your university name! Title this email “Final Case A - Team [Number]”.

4 Guidelines for using ChatGPT and other AI tools

- Allowed Usage:
ChatGPT can be used to assist with coding tasks, including debugging, optimizing, and understanding programming concepts. You may use ChatGPT to generate code snippets, explore different coding approaches, and troubleshoot errors.
- Not Allowed Usage:
ChatGPT cannot be used to write your report, including explanations, analysis, or conclusions. All written content in your report must be your own work, based on your understanding and research.

5 Feature names and descriptions

`begin_date` - Begin date of redispach
`end_date` - End date of redispach
`operator` - Operator
`direction` - Direction
`mean_load` - Average load
`max_load` - Maximum load
`total_load` - Total load
`n_concurrent_redispatches` - Number of concurrent redispatches
`brent_oil_price_open` - Opening price of Brent oil, USD/Bbl
`brent_oil_price_close` - Closing price of Brent oil, USD/Bbl
`brent_oil_price_high` - Highest price of Brent oil during a trade session, USD/Bbl
`brent_oil_price_low` - Lowest price of Brent oil during a trade session, USD/Bbl
`carbon_emissions_futures_price` - Clothing price of carbon emission futures, USD/K
`carbon_emissions_futures_open` - Opening price of carbon emission futures, USD/K
`carbon_emissions_futures_high` - Highest price of carbon emission futures during a trade session, USD/K
`carbon_emissions_futures_low` - Lowest price of carbon emission futures during a trade session, USD/K
`carbon_emissions_futures_vol` - Trade volume of carbon emission futures pro trade session, in thousands of units
`electricity_day_ahead_price_ch` - Day-ahead price of electricity in Switzerland, USD/MWh
`electricity_day_ahead_price_cz` - Day-ahead price of electricity in Chech Republic, USD/MWh,
`electricity_day_ahead_price_be` - Day-ahead price of electricity in Belgium, USD/MWh

electricity_day_ahead_price_nl - Day-ahead price of electricity in the Netherlands, USD/MWh
 electricity_day_ahead_price_pl - Day-ahead price of electricity in Poland, USD/MWh
 electricity_day_ahead_price_at - Day-ahead price of electricity in Austria, USD/MWh
 electricity_day_ahead_price_de_lu - Day-ahead price of electricity in Germany and Luxembourg, USD/MWh
 natural_gas_price_open - Opening price of natural gas, USD/MMBtu
 natural_gas_price_close - Closing price of natural gas, USD/MMBtu
 natural_gas_price_high - Highest price of natural gas during a trade session, USD/MMBtu
 natural_gas_price_low - Lowest price of natural gas during a trade session, USD/MMBtu
 sunshine_duration - Average sunshine duration per TSO per hour, in minutes
 air_temperature - Average air temperature at a height of 2 meters per TSO per hour, in Celcius
 wind_speed - Average wind speed per TSO per hour, in m/s
 day - Day
 hour - Hour
 month - Month
 is_workday - Workday dummy
 is_weekend - Weekend dummy
 is_holiday - Holiday dummy
 consumption_forecast_grid_load_mwh - Day-ahead forecast of grid load for Germany per hour, MWh
 consumption_forecast_residual_load_mwh - Day-ahead forecast of residual load for Germany per hour, MWh
 day_ahead_auktion - Day-ahead auction price of CO2 emission certificate, USD/tCO2
 co2_emissions_certificate_auction_de - Price of CO2 emission certificate in Germany, USD/tCO2
 co2_emissions_certificate_auction_eu - Price of CO2 emission certificate in EU, USD/tCO2
 production_forecast_total_mwh - Day-ahead forecast of total electricity production in Germany per hour, MWh
 production_forecast_photovoltaics_and_wid_mwh - Day-ahead forecast of wind and photovoltaic electricity generation in Germany per hour, MWh
 production_forecast_wind_offshore_mwh - Day-ahead forecast of wind and photovoltaic electricity generation in Germany per hour, MWh
 production_forecast_wind_onshore_mwh - Day-ahead forecast of wind onshore electricity generation in Germany per hour, MWh
 production_forecast_photovoltaics_mwh - Day-ahead forecast of photovoltaic electricity generation in Germany per hour, MWh
 production_forecast_other_mwh - Day-ahead forecast of electricity generated from conventional sources in Germany per hour, MWh

mean_sunshine_duration_bayern - Average sunshine duration in the German federal state per hour, in minutes
min_sunshine_duration_bayern - Minimum sunshine duration in the German federal state per hour, in minutes
max_sunshine_duration_bayern - Maximum sunshine duration in the German federal state per hour, in minutes
mean_sunshine_duration_bremen - - Average sunshine duration in the German federal state per hour, in minutes
min_sunshine_duration_bremen - Minimum sunshine duration in the German federal state per hour, in minutes
max_sunshine_duration_bremen - Maximum sunshine duration in the German federal state per hour, in minutes
mean_sunshine_duration_hessen - Average sunshine duration in the German federal state per hour, in minutes
min_sunshine_duration_hessen - Minimum sunshine duration in the German federal state per hour, in minutes
max_sunshine_duration_hessen - Maximum sunshine duration in the German federal state per hour, in minutes
mean_sunshine_duration_niedersachsen - Average sunshine duration in the German federal state per hour, in minutes
min_sunshine_duration_niedersachsen - Minimum sunshine duration in the German federal state per hour, in minutes
max_sunshine_duration_niedersachsen - Maximum sunshine duration in the German federal state per hour, in minutes
mean_sunshine_duration_schleswig_holstein - Average sunshine duration in the German federal state per hour, in minutes
min_sunshine_duration_schleswig_holstein - Minimum sunshine duration in the German federal state per hour, in minutes
max_sunshine_duration_schleswig_holstein - Maximum sunshine duration in the German federal state per hour, in minutes
mean_wind_velocity_bayern - Maximum sunshine duration in the German federal state per hour, in minutes
min_wind_velocity_bayern - Minimum wind speed per hour in the German federal state, in m/s
max_wind_velocity_bayern - Minimum wind speed per hour in the German federal state, in m/s
mean_wind_velocity_bremen - Maximum sunshine duration in the German federal state per hour, in minutes
min_wind_velocity_bremen - Minimum wind speed per hour in the German federal state, in m/s
max_wind_velocity_bremen - Minimum wind speed per hour in the German federal state, in m/s
mean_wind_velocity_hessen - Maximum sunshine duration in the German federal state per hour, in minutes
min_wind_velocity_hessen - Minimum wind speed per hour in the German federal state, in m/s

`max_wind_velocity_hessen` - Minimum wind speed per hour in the German federal state, in m/s
`mean_wind_velocity_niedersachsen` - Maximum sunshine duration in the German federal state per hour, in minutes
`min_wind_velocity_niedersachsen` - Minimum wind speed per hour in the German federal state, in m/s
`max_wind_velocity_niedersachsen` - Minimum wind speed per hour in the German federal state, in m/s
`mean_wind_velocity_schleswig_holstein` - Maximum sunshine duration in the German federal state per hour, in minutes
`min_wind_velocity_schleswig_holstein` - Minimum wind speed per hour in the German federal state, in m/s
`max_wind_velocity_schleswig_holstein` - Minimum wind speed per hour in the German federal state, in m/s

6 Literature

Billault-Chaumartin, Florence, Manuel Eising, and Agustín Motte-Cortés. "Redispatch patterns of energy market and reserve power plants in Germany." 2020 17th International Conference on the European Energy Market (EEM). IEEE, 2020.

Gürses-Tran, Gonca, Hendrik Flamme, and Antonello Monti. "Probabilistic load forecasting for day-ahead congestion mitigation." 2020 International Conference on Probabilistic Methods Applied to Power Systems (PMAPS). IEEE, 2020.

Titz, Maurizio, Sebastian Pütz, and Dirk Witthaut. "Identifying drivers and mitigators for congestion and redispatch in the German electric power system with explainable AI." *Applied Energy* 356 (2024): 122351.

Zhou, Qun, Leigh Tesfatsion, and Chen-Ching Liu. "Short-term congestion forecasting in wholesale power markets." *IEEE Transactions on Power Systems* 26.4 (2011): 2185-2196.